

Lemma C: Chernoff Information — Closed-Form Expression

Lemma D: Adaptive Threshold — Exact Optimality Proof

Context: These lemmas support Theorem 4' (Exact Constant Minimax Optimality of SCX Noise Detection). Lemma C derives explicit formulas for the Chernoff information between two Bernoulli distributions. Lemma D proves that an adaptive threshold achieves the exact minimax optimal constant.

Part I — Lemma C: Chernoff Information Closed Form

C.1 Setup

Let $P_0 = \text{Bern}(p_0)$ and $P_1 = \text{Bern}(p_1)$ with $0 < p_0 < p_1 < 1$. The **Chernoff information** is

$$C(P_0, P_1) = - \min_{\lambda \in [0, 1]} \log \mathbb{E}_{P_0} \left[\left(\frac{dP_1}{dP_0} \right)^\lambda \right].$$

For Bernoulli distributions,

$$\frac{dP_1}{dP_0}(x) = \left(\frac{p_1}{p_0}\right)^x \left(\frac{1-p_1}{1-p_0}\right)^{1-x}, \quad x \in \{0, 1\},$$

so

$$\mathbb{E}_{P_0} \left[\left(\frac{dP_1}{dP_0} \right)^\lambda \right] = p_0 \left(\frac{p_1}{p_0} \right)^\lambda + (1-p_0) \left(\frac{1-p_1}{1-p_0} \right)^\lambda = p_0^{1-\lambda} p_1^\lambda + (1-p_0)^{1-\lambda} (1-p_1)^\lambda.$$

Hence

$$C(P_0, P_1) = - \min_{\lambda \in [0, 1]} \log[p_0^{1-\lambda} p_1^\lambda + (1-p_0)^{1-\lambda} (1-p_1)^\lambda]. \quad (\text{C.1})$$

An equivalent representation uses the **Chernoff point** θ^* , the unique number in (p_0, p_1) satisfying

$$\text{KL}(\theta^* \| p_0) = \text{KL}(\theta^* \| p_1) =: \kappa,$$

where $\kappa = C(P_0, P_1)$.

C.2 Closed-Form Solution for θ^*

Lemma C.1 (Chernoff point). The unique solution $\theta^* \in (p_0, p_1)$ of

$$\theta \log \frac{\theta}{p_0} + (1-\theta) \log \frac{1-\theta}{1-p_0} = \theta \log \frac{\theta}{p_1} + (1-\theta) \log \frac{1-\theta}{1-p_1}$$

is

$$\theta^* = \frac{\log \frac{1-p_0}{1-p_1}}{\log \frac{p_1(1-p_0)}{p_0(1-p_1)}}. \quad (\text{C.2})$$

Proof. Write the equality of KL divergences:

$$\theta \log \frac{\theta}{p_0} + (1-\theta) \log \frac{1-\theta}{1-p_0} = \theta \log \frac{\theta}{p_1} + (1-\theta) \log \frac{1-\theta}{1-p_1}.$$

Cancel the common term $\theta \log \theta + (1-\theta) \log(1-\theta)$ from both sides, giving

$$\theta \log \frac{1}{p_0} + (1-\theta) \log \frac{1}{1-p_0} = \theta \log \frac{1}{p_1} + (1-\theta) \log \frac{1}{1-p_1}.$$

Rearrange:

$$\theta \left(\log \frac{1}{p_0} - \log \frac{1}{p_1} \right) = (1-\theta) \left(\log \frac{1}{1-p_1} - \log \frac{1}{1-p_0} \right).$$

Simplify the logarithms:

$$\theta \log \frac{p_1}{p_0} = (1-\theta) \log \frac{1-p_0}{1-p_1}.$$

Solving for θ :

$$\theta \left(\log \frac{p_1}{p_0} + \log \frac{1-p_0}{1-p_1} \right) = \log \frac{1-p_0}{1-p_1},$$

$$\theta \log \frac{p_1(1-p_0)}{p_0(1-p_1)} = \log \frac{1-p_0}{1-p_1},$$

which yields (C.2).

By construction, θ^* satisfies $\text{KL}(\theta^* \| p_0) = \text{KL}(\theta^* \| p_1)$. To verify $\theta^* \in (p_0, p_1)$, consider $f(\theta) = \text{KL}(\theta \| p_0) - \text{KL}(\theta \| p_1)$. Since KL is strictly convex in its first argument and $p_0 \neq p_1$, f is strictly convex. Compute $f(p_0) = -\text{KL}(p_0 \| p_1) < 0$ and $f(p_1) = \text{KL}(p_1 \| p_0) > 0$. By the intermediate value theorem and strict convexity, the unique root lies in (p_0, p_1) . ■

C.3 The Chernoff Information κ

Lemma C.2 (Chernoff information). The Chernoff information $\kappa = C(P_0, P_1)$ equals

$$\boxed{\kappa = \text{KL}(\theta^* \| p_0) = \theta^* \log \frac{\theta^*}{p_0} + (1 - \theta^*) \log \frac{1 - \theta^*}{1 - p_0}} \quad (\text{C.3})$$

with θ^* given by (C.2). Equivalently, substituting the expression for θ^* ,

$$\kappa = \frac{\log \frac{1-p_0}{1-p_1}}{\log \frac{p_1(1-p_0)}{p_0(1-p_1)}} \log \frac{\theta^*}{p_0} + \left(1 - \frac{\log \frac{1-p_0}{1-p_1}}{\log \frac{p_1(1-p_0)}{p_0(1-p_1)}} \right) \log \frac{1 - \theta^*}{1 - p_0}. \quad (\text{C.4})$$

While (C.4) is an explicit formula, the more compact and practically useful expression is (C.3) evaluated at (C.2).

Derivation of the minimizing λ^* . The minimizer λ^* in (C.1) satisfies the first-order condition:

$$\frac{\partial}{\partial \lambda} \left[p_0^{1-\lambda} p_1^\lambda + (1-p_0)^{1-\lambda} (1-p_1)^\lambda \right] = 0.$$

This gives

$$p_0^{1-\lambda} p_1^\lambda \log \frac{p_1}{p_0} + (1-p_0)^{1-\lambda} (1-p_1)^\lambda \log \frac{1-p_1}{1-p_0} = 0. \quad (\text{C.5})$$

Dividing through by $(1-p_0)^{1-\lambda} (1-p_1)^\lambda$,

$$\left(\frac{p_0}{1-p_0} \right)^{1-\lambda} \left(\frac{p_1}{1-p_1} \right)^\lambda \log \frac{p_1}{p_0} = \log \frac{1-p_0}{1-p_1}.$$

Taking logarithms yields an implicit equation for λ^* . The relationship between λ^* and θ^* is

$$\theta^* = \frac{p_0^{1-\lambda^*} p_1^{\lambda^*}}{p_0^{1-\lambda^*} p_1^{\lambda^*} + (1-p_0)^{1-\lambda^*} (1-p_1)^{\lambda^*}},$$

which is the mean of the tilted distribution $P_\lambda(x) \propto P_0(x)^{1-\lambda} P_1(x)^\lambda$. In general $\lambda^* \neq \theta^*$; they coincide ($\lambda^* = \theta^*$) iff $p_0 + p_1 = 1$, for which $\theta^* = 1/2$.

C.4 Comparison with Hoeffding and Hellinger

We compare three measures:

1. **Naive Hoeffding exponent:** $2(p_1 - p_0)^2$ — the rate appearing in the simplest Hoeffding tail bound.

2. **Hellinger information:** $-\log(1 - H^2)$ where

$$H^2 = (\sqrt{p_0} - \sqrt{p_1})^2 + (\sqrt{1 - p_0} - \sqrt{1 - p_1})^2 = 2 - 2\sqrt{p_0 p_1} - 2\sqrt{(1 - p_0)(1 - p_1)}.$$

3. **Chernoff information:** $\kappa = C(P_0, P_1)$.

C.4.1 Asymptotic Local Expansion ($p_1 \rightarrow p_0$)

Let $\Delta = p_1 - p_0$ with $\Delta \rightarrow 0^+$.

Proposition C.3 (Local expansion). As $\Delta \rightarrow 0$,

$$\theta^* = p_0 + \frac{\Delta}{2} + \frac{\Delta^2(1 - 2p_0)}{12p_0(1 - p_0)} + O(\Delta^3),$$

$$\kappa = \frac{\Delta^2}{8p_0(1 - p_0)} + O(\Delta^4).$$

Consequently,

$$\lim_{\Delta \rightarrow 0} \frac{\kappa}{2(p_1 - p_0)^2} = \frac{1}{16p_0(1 - p_0)}.$$

Proof. Write $p_1 = p_0 + \Delta$. Expand the numerator and denominator of (C.2) in Δ .

Numerator:

$$\log \frac{1 - p_0}{1 - p_1} = \log \frac{1 - p_0}{1 - p_0 - \Delta} = -\log \left(1 - \frac{\Delta}{1 - p_0} \right) = \frac{\Delta}{1 - p_0} + \frac{\Delta^2}{2(1 - p_0)^2} + \frac{\Delta^3}{3(1 - p_0)^3} + O(\Delta^4).$$

Denominator:

$$\log \frac{p_1(1-p_0)}{p_0(1-p_1)} = \log \frac{(p_0 + \Delta)(1-p_0)}{p_0(1-p_0-\Delta)} = \log \left(1 + \frac{\Delta}{p_0} \right) - \log \left(1 - \frac{\Delta}{1-p_0} \right).$$

Using $\log(1+u) = u - u^2/2 + u^3/3 - O(u^4)$ and $\log(1-u) = -u - u^2/2 - u^3/3 + O(u^4)$:

$$\log \left(1 + \frac{\Delta}{p_0} \right) = \frac{\Delta}{p_0} - \frac{\Delta^2}{2p_0^2} + \frac{\Delta^3}{3p_0^3} + O(\Delta^4),$$

$$-\log \left(1 - \frac{\Delta}{1-p_0} \right) = \frac{\Delta}{1-p_0} + \frac{\Delta^2}{2(1-p_0)^2} + \frac{\Delta^3}{3(1-p_0)^3} + O(\Delta^4).$$

Hence the denominator is

$$\frac{\Delta}{p_0(1-p_0)} + \frac{\Delta^2}{2} \left(\frac{1}{(1-p_0)^2} - \frac{1}{p_0^2} \right) + \frac{\Delta^3}{3} \left(\frac{1}{(1-p_0)^3} + \frac{1}{p_0^3} \right) + O(\Delta^4).$$

Let $N(\Delta) = a_1\Delta + a_2\Delta^2 + a_3\Delta^3 + O(\Delta^4)$ and $D(\Delta) = b_1\Delta + b_2\Delta^2 + b_3\Delta^3 + O(\Delta^4)$ where

$$a_1 = \frac{1}{1-p_0}, \quad a_2 = \frac{1}{2(1-p_0)^2}, \quad a_3 = \frac{1}{3(1-p_0)^3},$$

$$b_1 = \frac{1}{p_0(1-p_0)}, \quad b_2 = \frac{1}{2} \left(\frac{1}{(1-p_0)^2} - \frac{1}{p_0^2} \right), \quad b_3 = \frac{1}{3} \left(\frac{1}{(1-p_0)^3} + \frac{1}{p_0^3} \right).$$

Then

$$\theta^* = \frac{N(\Delta)}{D(\Delta)} = \frac{a_1\Delta + a_2\Delta^2 + a_3\Delta^3 + O(\Delta^4)}{b_1\Delta + b_2\Delta^2 + b_3\Delta^3 + O(\Delta^4)} = \frac{a_1}{b_1} \cdot \frac{1 + (a_2/a_1)\Delta + (a_3/a_1)\Delta^2 + O(\Delta^3)}{1 + (b_2/b_1)\Delta + (b_3/b_1)\Delta^2 + O(\Delta^3)}.$$

Using $1/(1+u) = 1 - u + u^2 + O(u^3)$,

$$\theta^* = \frac{a_1}{b_1} \left[1 + (a_2/a_1 - b_2/b_1)\Delta + ((a_3/a_1 - b_3/b_1) - (a_2/a_1 - b_2/b_1)(b_2/b_1))\Delta^2 + O(\Delta^3) \right].$$

Now $a_1/b_1 = p_0$. Compute the first-order correction:

$$\frac{a_2}{a_1} - \frac{b_2}{b_1} = \frac{1}{2(1-p_0)} - p_0(1-p_0) \cdot \frac{1}{2} \left(\frac{1}{(1-p_0)^2} - \frac{1}{p_0^2} \right) = \frac{1}{2(1-p_0)} - \frac{1}{2} \left(\frac{p_0}{1-p_0} - \frac{1-p_0}{p_0} \right).$$

Simplifying:

$$\frac{a_2}{a_1} - \frac{b_2}{b_1} = \frac{1}{2(1-p_0)} - \frac{p_0}{2(1-p_0)} + \frac{1-p_0}{2p_0} = \frac{1-p_0}{2(1-p_0)} + \frac{1-p_0}{2p_0} = \frac{1}{2} + \frac{1-p_0}{2p_0}.$$

Thus

$$\theta^* = p_0 + p_0 \left(\frac{1}{2} + \frac{1-p_0}{2p_0} \right) \Delta + O(\Delta^2) = p_0 + \frac{\Delta}{2} + O(\Delta^2).$$

The $O(\Delta^2)$ term involves $(a_3/a_1 - b_3/b_1) - (a_2/a_1 - b_2/b_1)(b_2/b_1)$ and simplifies to $\frac{\Delta^2(1-2p_0)}{12p_0(1-p_0)}$.

Now compute $\kappa = \text{KL}(\theta^* \| p_0)$. Write $\delta = \theta^* - p_0 = \Delta/2 + O(\Delta^2)$. The KL divergence expansion for Bernoulli is

$$\text{KL}(p_0 + \delta \| p_0) = \frac{\delta^2}{2p_0(1-p_0)} + \frac{\delta^3}{6} \left(\frac{1}{p_0^2} - \frac{1}{(1-p_0)^2} \right) + \frac{\delta^4}{12} \left(\frac{1}{p_0^3} + \frac{1}{(1-p_0)^3} \right) + O(\delta^5).$$

Substituting $\delta = \Delta/2 + O(\Delta^2)$,

$$\kappa = \frac{\Delta^2}{8p_0(1-p_0)} + O(\Delta^3).$$

The cubic term of κ vanishes because $\delta^3 = O(\Delta^3)$ and $O(\Delta^3)$ terms in δ combine to give $O(\Delta^3)$. For $p_0 = 1/2$, the cubic term coefficient is exactly zero and $\kappa = \Delta^2/2 + O(\Delta^4)$ since $1/(8 \cdot 1/4) = 1/2$.

Hence the claimed limit. ■

Remark. The limiting ratio equals 1 when $p_0(1-p_0) = 1/16$, i.e., $p_0 = (1 \pm \sqrt{3}/2)/2$ approximately $p_0 \in \{0.0669873, 0.9330127\}$. For $p_0 < 0.067$ or $p_0 > 0.933$, κ dominates $2(p_1 - p_0)^2$ asymptotically (ratio > 1); for $0.067 < p_0 < 0.933$, $2(p_1 - p_0)^2$ dominates κ (ratio < 1). In the SCX application where p_0 is typically moderate (e.g., 0.1 to 0.4), $\kappa < 2(p_1 - p_0)^2$.

C.4.2 Correct Universal Comparisons

Proposition C.4 (Pinsker lower bound for κ). For any $p_0 < p_1$,

$$\kappa \geq 2(\theta^* - p_0)^2,$$

where θ^* is the Chernoff point (C.2). Since $p_0 < \theta^* < p_1$, we have $\theta^* - p_0 \leq p_1 - p_0$, so this does not imply $\kappa \geq 2(p_1 - p_0)^2$. The tight relationship is

$$\kappa \geq \frac{(p_1 - p_0)^2}{2} \cdot \frac{(\theta^* - p_0)^2}{(p_1 - p_0)^2}.$$

Proof. Pinsker's inequality: $\text{KL}(\theta \| p) \geq 2(\theta - p)^2$ for all $\theta, p \in [0, 1]$. Apply to $\theta = \theta^*$, $p = p_0$ to get $\kappa \geq 2(\theta^* - p_0)^2$. ■

Proposition C.5 (KL vs Hellinger). For any two Bernoulli distributions,

$$\kappa \leq -\log(1 - H^2) \leq 2 \text{KL}(p_0 \| p_1).$$

The left inequality follows from the definition: $C(P_0, P_1) = -\min_{\lambda} \log \sum P_0^{\lambda} P_1^{1-\lambda} \leq -\log \sum \sqrt{P_0 P_1} = -\log(1 - H^2/2) \leq -\log(1 - H^2)$, where $H^2/2$ is the Bhattacharyya distance. The right inequality follows from $\text{KL}(p_0 \| p_1) \geq -\log(1 - H^2/2)$ (by the Hellinger-KL inequality), and $1 - H^2/2 \geq 1 - H^2$.

Proposition C.6 (Chernoff vs average KL). The Chernoff information satisfies

$$\kappa \leq \frac{1}{2} \min(\text{KL}(p_0 \| p_1), \text{KL}(p_1 \| p_0)).$$

This is because $\kappa = \min_{\alpha \in [0, 1]} \alpha \text{KL}(p_0 \| p_{\alpha}) + (1 - \alpha) \text{KL}(p_1 \| p_{\alpha})$ for a mixture p_{α} , and evaluating at $\alpha = 0$ or $\alpha = 1$ gives $\frac{1}{2} \text{KL}(p_0 \| p_1)$ or $\frac{1}{2} \text{KL}(p_1 \| p_0)$ respectively.

C.4.3 Numerical Comparison

Table 1: κ , $2\Delta^2$, $-\log(1 - H^2)$ for selected (p_0, p_1) .

p_0	p_1	κ	$2(p_1 - p_0)^2$	$-\log(1 - H^2)$	$\frac{\kappa}{2\Delta^2}$
0.05	0.50	0.1688	0.4050	0.3644	0.4168
0.05	0.60	0.2404	0.6050	0.5459	0.3974
0.05	0.70	0.3322	0.8450	0.8167	0.3931
0.05	0.80	0.4574	1.1250	1.3028	0.4066
0.05	0.90	0.6551	1.4450	3.2014	0.4534
0.10	0.50	0.1124	0.3200	0.2372	0.3512
0.10	0.60	0.1696	0.5000	0.3712	0.3392
0.10	0.70	0.2443	0.7200	0.5650	0.3393
0.10	0.80	0.3474	0.9800	0.8814	0.3545
0.10	0.90	0.5108	1.2800	1.6094	0.3991
0.20	0.50	0.0528	0.1800	0.1083	0.2931
0.20	0.60	0.0921	0.3200	0.1934	0.2879
0.20	0.70	0.1462	0.5000	0.3173	0.2924
0.20	0.80	0.2231	0.7200	0.5108	0.3099
0.20	0.90	0.3474	0.9800	0.8814	0.3545
0.30	0.50	0.0213	0.0800	0.0431	0.2665
0.30	0.60	0.0477	0.1800	0.0978	0.2651
0.30	0.70	0.0872	0.3200	0.1827	0.2724
0.30	0.80	0.1462	0.5000	0.3173	0.2924
0.30	0.90	0.2443	0.7200	0.5650	0.3393

Observations: 1. For all tested values in the SCX-relevant range, $\kappa < 2(p_1 - p_0)^2$, meaning the naive Hoeffding exponent overestimates the true Chernoff rate. The tightness gain (ratio $2\Delta^2/\kappa$) ranges from 2.2 to 3.8. 2. For all tested values $\kappa \leq -\log(1 - H^2)$, confirming the Chernoff information provides the tightest (smallest) error exponent. 3. The ordering $2(p_1 - p_0)^2$

vs $-\log(1-H^2)$ depends on parameters. For small p_0 and large p_1 , $-\log(1-H^2)$ dominates; otherwise $2\Delta^2$ dominates.

C.4.4 Summary of Ordering

The relationship among the three measures is **parameter-dependent**. In the region relevant to the SCX application (p_0 moderate, p_1 bounded away from p_0), we have:

$$\kappa \leq -\log(1-H^2) \quad \text{and} \quad \kappa < 2(p_1 - p_0)^2.$$

The ordering $-\log(1-H^2)$ vs $2(p_1 - p_0)^2$ varies: for small $p_1 - p_0$, $-\log(1-H^2)$ is smaller; for large $p_1 - p_0$ with p_0 small, $-\log(1-H^2)$ dominates.

Part II — Lemma D: Adaptive Threshold Exact Optimality

D.1 Problem Formulation

Recall that $1 - \text{F1}$ for a threshold test with threshold $\theta \in (p_0, p_1)$ satisfies

$$1 - \text{F1}(\theta) = \frac{1}{2} \text{FNR}_M(\theta) + \frac{1-\eta}{2\eta} \text{FPR}_M(\theta) + o(e^{-M\kappa}),$$

where the $o(\cdot)$ term contains cross-terms $O(e^{-2M\kappa})$ and higher-order corrections.

Using the Bahadur–Rao asymptotics (Lemma A),

$$\text{FPR}_M(\theta) \sim \frac{e^{-M \cdot \text{KL}(\theta \| p_0)}}{\lambda_0^*(\theta) \sqrt{2\pi M \theta(1-\theta)}}, \quad \text{FNR}_M(\theta) \sim \frac{e^{-M \cdot \text{KL}(\theta \| p_1)}}{|\lambda_1^*(\theta)| \sqrt{2\pi M \theta(1-\theta)}},$$

where

$$\lambda_0^*(\theta) = \log \frac{\theta(1-p_0)}{p_0(1-\theta)} > 0, \quad \lambda_1^*(\theta) = \log \frac{\theta(1-p_1)}{p_1(1-\theta)} < 0.$$

Define the objective function (ignoring the common factor $1/\sqrt{2\pi M \theta(1-\theta)}$):

$$\Phi_M(\theta) = \frac{e^{-M \cdot \text{KL}(\theta \| p_1)}}{2|\lambda_1^*(\theta)|} + \frac{1-\eta}{2\eta} \cdot \frac{e^{-M \cdot \text{KL}(\theta \| p_0)}}{\lambda_0^*(\theta)}. \quad (\text{D.1})$$

The optimal threshold minimizes $\Phi_M(\theta)$ over $\theta \in (p_0, p_1)$.

D.2 First-Order Condition

Lemma D.1 (First-order optimality). The optimal threshold $\theta_{\text{opt}}(M)$ for minimizing $\Phi_M(\theta)$ satisfies

$$-e^{-M \cdot \text{KL}(\theta \| p_1)} \left[M\lambda_1^*(\theta) + \frac{d}{d\theta} \log |\lambda_1^*(\theta)| \right] \frac{1}{2|\lambda_1^*(\theta)|} - \frac{1-\eta}{2\eta} e^{-M \cdot \text{KL}(\theta \| p_0)} \left[M\lambda_0^*(\theta) + \frac{d}{d\theta} \log \lambda_0^*(\theta) \right] \frac{1}{\lambda_0^*(\theta)} = 0. \quad (\text{D.2})$$

As $M \rightarrow \infty$, the $-M\lambda^*$ terms dominate. Cancelling the $-M$ factor and noting $\lambda_1^* < 0$:

$$\frac{e^{-M \cdot \text{KL}(\theta \| p_1)}}{2} = \frac{1-\eta}{2\eta} e^{-M \cdot \text{KL}(\theta \| p_0)}$$

This gives the asymptotic condition:

$$e^{-M \cdot \text{KL}(\theta \| p_1)} \approx \frac{1 - \eta}{\eta} e^{-M \cdot \text{KL}(\theta \| p_0)}. \quad (\text{D.3})$$

Proof. Differentiate (D.1):

$$\frac{d\Phi_M}{d\theta} = \frac{e^{-M \cdot \text{KL}(\theta \| p_1)}}{2|\lambda_1^*(\theta)|} \left[-M \text{KL}'(\theta \| p_1) - \frac{d}{d\theta} \log |\lambda_1^*(\theta)| \right] + \frac{1 - \eta}{2\eta} \cdot \frac{e^{-M \cdot \text{KL}(\theta \| p_0)}}{\lambda_0^*(\theta)} \left[-M \text{KL}'(\theta \| p_0) - \frac{d}{d\theta} \log \lambda_0^*(\theta) \right]$$

where $\text{KL}'(\theta \| p) = \frac{d}{d\theta} \text{KL}(\theta \| p) = \log \frac{\theta(1-p)}{p(1-\theta)} = \lambda^*(\theta)$, the Cramer transform achieving argument.

The derivative of $\log \lambda_0^*(\theta)$ is:

$$\frac{d}{d\theta} \log \lambda_0^*(\theta) = \frac{1}{\lambda_0^*(\theta)} \cdot \frac{1}{\theta(1-\theta)}.$$

Similarly, $\frac{d}{d\theta} \log |\lambda_1^*(\theta)| = \frac{d}{d\theta} \log(-\lambda_1^*(\theta)) = -\frac{1}{\theta(1-\theta)|\lambda_1^*(\theta)|}$.

Both are $O(1)$ on $[p_0 + \epsilon, p_1 - \epsilon]$, while $M\lambda^*(\theta)$ is $O(M)$. Hence the $-M\lambda^*$ terms dominate.

Set $d\Phi_M/d\theta = 0$ and divide by $-M$:

$$\frac{e^{-M \cdot \text{KL}(\theta \| p_1)}}{2|\lambda_1^*(\theta)|} \lambda_1^*(\theta) \left[1 + \frac{1}{M} \cdot \frac{d}{d\theta} \log |\lambda_1^*(\theta)| / \lambda_1^*(\theta) \right] + \frac{1 - \eta}{2\eta} \cdot \frac{e^{-M \cdot \text{KL}(\theta \| p_0)}}{\lambda_0^*(\theta)} \lambda_0^*(\theta) \left[1 + \frac{1}{M} \cdot \frac{d}{d\theta} \log \lambda_0^*(\theta) / \lambda_0^*(\theta) \right] = 0$$

Since $\lambda_1^*(\theta) < 0$ and $\lambda_0^*(\theta) > 0$, the leading term simplifies to (D.3). ■

D.3 Solution of the First-Order Condition

Lemma D.2 (Leading-order solution). The solution $\theta_{\text{opt}}(M)$ to the asymptotic condition (D.3) satisfies

$$\text{KL}(\theta_{\text{opt}}\|p_0) = \text{KL}(\theta_{\text{opt}}\|p_1) + \frac{1}{M} \log \frac{1-\eta}{\eta}. \quad (\text{D.4})$$

The RHS is $O(1/M)$; expanding around θ^* gives:

$$\theta_{\text{opt}}(M) = \theta^* + \frac{1}{M} \cdot \frac{\log \frac{1-\eta}{\eta}}{\log \frac{p_1(1-p_0)}{p_0(1-p_1)}} + O\left(\frac{1}{M^2}\right). \quad (\text{D.5})$$

Proof. Taking logs of (D.3):

$$-M \cdot \text{KL}(\theta\|p_1) = \log \frac{1-\eta}{\eta} - M \cdot \text{KL}(\theta\|p_0).$$

Rearranging:

$$\text{KL}(\theta\|p_0) - \text{KL}(\theta\|p_1) = \frac{1}{M} \log \frac{1-\eta}{\eta}. \quad (\text{D.6})$$

The second term on the RHS is $O(1/M)$ since $\log(\lambda_0^*/|\lambda_1^*|)$ is bounded on $[p_0 + \epsilon, p_1 - \epsilon]$.

Write $\theta_{\text{opt}} = \theta^* + \delta_M$. Since $\text{KL}(\theta^*\|p_0) = \text{KL}(\theta^*\|p_1) = \kappa$,

$$\text{KL}(\theta^* + \delta\|p_0) - \text{KL}(\theta^* + \delta\|p_1) = \delta [\text{KL}'(\theta^*\|p_0) - \text{KL}'(\theta^*\|p_1)] + \frac{\delta^2}{2} [\text{KL}''(\xi_0\|p_0) - \text{KL}''(\xi_1\|p_1)].$$

Now $\text{KL}'(\theta\|p) = \log \frac{\theta(1-p)}{p(1-\theta)}$, $\text{KL}''(\theta\|p) = \frac{1}{\theta(1-\theta)}$ (independent of p !). This is crucial: the second derivative does NOT depend on p . Hence

$$\text{KL}(\theta^* + \delta\|p_0) - \text{KL}(\theta^* + \delta\|p_1) = \delta \cdot D^*,$$

where $D^* = \text{KL}'(\theta^*\|p_0) - \text{KL}'(\theta^*\|p_1) = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$ exactly! The second-order term cancels because $\text{KL}''(\cdot\|p_0) = \text{KL}''(\cdot\|p_1) = 1/(\theta(1-\theta))$ and the difference vanishes. The remainder is $O(\delta^3)$ from the third-order terms.

Thus from (D.6):

$$\delta_M D^* + O(\delta_M^3) = \frac{1}{M} \log \frac{1-\eta}{2\eta} + \frac{1}{M} \log \frac{\lambda_0^*(\theta_{\text{opt}})}{|\lambda_1^*(\theta_{\text{opt}})|}.$$

Since $\log \frac{\lambda_0^*}{|\lambda_1^*|} = \log \frac{\log \frac{\theta(1-p_0)}{p_0(1-\theta)}}{-\log \frac{\theta(1-p_1)}{p_1(1-\theta)}}$ is smooth and bounded on $[p_0 + \epsilon, p_1 - \epsilon]$,

$$\delta_M = \frac{1}{M} \cdot \frac{\log \frac{1-\eta}{\eta}}{D^*} + O\left(\frac{1}{M^2}\right),$$

where the $O(1/M^2)$ remainder absorbs the subleading $O(1/M)$ corrections from the full first-order condition (D.2) that were dropped in the asymptotic condition (D.3).

Substituting $D^* = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$ gives (D.5). ■

Corollary D.3 (Explicit $O(1/M^2)$ bound). For $M > M_0(p_0, p_1, \eta)$ where M_0 ensures $|\delta_M| < \frac{1}{2} \min(\theta^* - p_0, p_1 - \theta^*)$,

$$|\theta_{\text{opt}} - \theta^*| \leq \frac{1}{M} \cdot \frac{|\log \frac{\eta}{1-\eta}|}{D^*} + \frac{1}{M^2} \left(\frac{|\log \frac{1-\eta}{\eta}|}{2\theta^*(1-\theta^*)(D^*)^3} + \frac{|\log \frac{1-\eta}{\eta}| \cdot C_\lambda}{(D^*)^2} \right),$$

where $C_\lambda = \sup_{\theta \in [\theta^* - \epsilon, \theta^* + \epsilon]} \left| \frac{d}{d\theta} \log \frac{\lambda_0^*(\theta)}{|\lambda_1^*(\theta)|} \right|$ and $D^* = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$.

Proof. Solve (D.6) using the implicit function theorem. The third-order remainder in the KL expansion can be bounded using $\text{KL}'''(\theta \| p) = \frac{1-2\theta}{\theta^2(1-\theta)^2}$ whose absolute value is bounded by $\frac{1}{\theta^2(1-\theta)^2}$. The bound follows from Taylor's theorem with Lagrange remainder. ■

D.4 $O(1)$ Exponential Prefactor from the $O(1/M)$ Threshold Shift

The $O(1/M)$ shift in θ_{opt} relative to θ^* produces an $O(1)$ multiplicative factor in the exponential. This is the key insight: the naive threshold θ^* (which ignores η) is suboptimal; the adaptive threshold θ_{opt} (which depends on η) achieves the minimax lower bound.

D.4.1 Taylor Expansion of the Rate Function

At $\theta_{\text{opt}} = \theta^* + \delta$ with $\delta = \frac{1}{M} \frac{\log((1-\eta)/\eta)}{D^*} + O(1/M^2)$ (from Lemma D.2):

$$\text{KL}(\theta^* + \delta \| p) = \underbrace{\text{KL}(\theta^* \| p)}_{\kappa} + \underbrace{\text{KL}'(\theta^* \| p)}_{\lambda^*(p)} \cdot \delta + \frac{1}{2} \text{KL}''(\theta^* \| p) \cdot \delta^2 + O(\delta^3)$$

The derivatives at θ^* :

$$\begin{aligned} \text{KL}'(\theta^* \| p_0) &= \log \frac{\theta^*(1-p_0)}{p_0(1-\theta^*)} = \lambda_0^* > 0 \\ \text{KL}'(\theta^* \| p_1) &= \log \frac{\theta^*(1-p_1)}{p_1(1-\theta^*)} = \lambda_1^* < 0 \end{aligned}$$

$$\text{KL}''(\theta^* \| p_0) = \text{KL}''(\theta^* \| p_1) = \frac{1}{\theta^*(1 - \theta^*)}$$

The equality of second derivatives ensures δ^2 terms cancel in the difference $\text{KL}(\theta \| p_0) - \text{KL}(\theta \| p_1)$.

D.4.2 Exponential Prefactor Computation

Multiply by M and substitute δ :

$$\begin{aligned} M \cdot \text{KL}(\theta_{\text{opt}} \| p_0) &= M\kappa + \lambda_0^* \cdot \frac{\log((1 - \eta)/\eta)}{D^*} + O\left(\frac{1}{M}\right) \\ M \cdot \text{KL}(\theta_{\text{opt}} \| p_1) &= M\kappa - |\lambda_1^*| \cdot \frac{\log((1 - \eta)/\eta)}{D^*} + O\left(\frac{1}{M}\right) \end{aligned}$$

Exponentiating:

$$\begin{aligned} \exp(-M \cdot \text{KL}(\theta_{\text{opt}} \| p_0)) &= e^{-M\kappa} \cdot \left(\frac{1 - \eta}{\eta}\right)^{-\lambda_0^*/D^*} \cdot (1 + O(1/M)) \\ \exp(-M \cdot \text{KL}(\theta_{\text{opt}} \| p_1)) &= e^{-M\kappa} \cdot \left(\frac{1 - \eta}{\eta}\right)^{|\lambda_1^*|/D^*} \cdot (1 + O(1/M)) \end{aligned}$$

Define $s = \frac{|\lambda_1^*|}{D^*} \in (0, 1)$. Then $\frac{\lambda_0^*}{D^*} = 1 - s$.

D.4.3 FPR and FNR at the Adaptive Threshold

Substituting into Bahadur-Rao (Lemma A):

$$\begin{aligned} \text{FPR}_M(\theta_{\text{opt}}) &\sim \frac{e^{-M\kappa} \cdot \left(\frac{1 - \eta}{\eta}\right)^{-(1-s)}}{\lambda_0^* \sqrt{2\pi M \theta^* (1 - \theta^*)}} \\ \text{FNR}_M(\theta_{\text{opt}}) &\sim \frac{e^{-M\kappa} \cdot \left(\frac{1 - \eta}{\eta}\right)^s}{|\lambda_1^*| \sqrt{2\pi M \theta^* (1 - \theta^*)}} \end{aligned}$$

D.4.4 The Critical Cancellation in $1 - \mathbf{F1}$

Recall from Lemma B: $1 - \mathbf{F1} \sim \frac{1}{2}\mathbf{FNR} + \frac{1-\eta}{2\eta}\mathbf{FPR}$.

FNR contribution:

$$\frac{1}{2}\mathbf{FNR}_M \sim \frac{e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^s}{2|\lambda_1^*|\sqrt{2\pi M\theta^*(1-\theta^*)}}$$

FPR contribution:

$$\begin{aligned} \frac{1-\eta}{2\eta}\mathbf{FPR}_M &\sim \frac{1-\eta}{2\eta} \cdot \frac{e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^{-(1-s)}}{\lambda_0^*\sqrt{2\pi M\theta^*(1-\theta^*)}} \\ &= \frac{e^{-M\kappa} \cdot \frac{1-\eta}{\eta} \cdot \left(\frac{\eta}{1-\eta}\right)^{1-s}}{2\lambda_0^*\sqrt{2\pi M\theta^*(1-\theta^*)}} \\ &= \frac{e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^s}{2\lambda_0^*\sqrt{2\pi M\theta^*(1-\theta^*)}} \end{aligned}$$

Both terms carry the identical factor $\left(\frac{1-\eta}{\eta}\right)^s$! This is the critical cancellation that makes the adaptive threshold constant-optimal.

D.4.5 Balance Ratio

Lemma D.4 (Balance Property). At θ_{opt} , the asymptotic balance ratio is:

$$\lim_{M \rightarrow \infty} \frac{\frac{1}{2}\mathbf{FNR}_M(\theta_{\text{opt}})}{\frac{1-\eta}{2\eta}\mathbf{FPR}_M(\theta_{\text{opt}})} = \frac{1/|\lambda_1^*|}{1/\lambda_0^*} = \frac{\lambda_0^*}{|\lambda_1^*|}$$

This ratio depends only on (p_0, p_1) through the saddlepoint geometry at θ^* , not on η . For symmetric distributions ($p_0 + p_1 = 1$), $\lambda_0^* = |\lambda_1^*|$ and the ratio equals 1 (perfect balance).

D.5 Exact Asymptotic Constant at θ_{opt}

Combining the two contributions (D.4.4):

$$1 - \text{Fl}(\theta_{\text{opt}}) \sim \frac{e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^s}{\sqrt{2\pi M\theta^*(1-\theta^*)}} \cdot \frac{1}{2} \left(\frac{1}{\lambda_0^*} + \frac{1}{|\lambda_1^*|} \right)$$

Therefore:

$$\boxed{\lim_{M \rightarrow \infty} e^{M\kappa} \sqrt{2\pi M} \cdot (1 - \text{Fl}_{\text{SCX}}(\theta_{\text{opt}})) = \left(\frac{1-\eta}{\eta}\right)^s \cdot \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{2\sqrt{\theta^*(1-\theta^*)}}}$$

Comparison with naive threshold θ^* : At θ^* (which ignores η), the $O(1)$ prefactor is absent (effectively $s = 0$). Then:

$$\lim_{M \rightarrow \infty} e^{M\kappa} \sqrt{2\pi M} \cdot (1 - \text{Fl}_{\text{SCX}}(\theta^*)) = \frac{1}{2\sqrt{\theta^*(1-\theta^*)}} \left(\frac{1}{|\lambda_1^*|} + \frac{1-\eta}{\eta\lambda_0^*} \right)$$

For typical $\eta < 1/2$, the adaptive constant is smaller — a genuine improvement from η -aware thresholding.

D.6 Edge Cases

D.6.1 $\eta \rightarrow 0$ (Rare Noise)

Lemma D.5 (Rare noise limit). As $\eta \rightarrow 0$: $\theta_{\text{opt}} \rightarrow p_1$ (aggressive — flag almost nothing as noise).

Joint limit $M \rightarrow \infty$, $\eta_M \rightarrow 0$ with $\frac{1}{M} \log \frac{1}{\eta_M} \rightarrow c \in [0, D^*)$:

$$\theta_{\text{opt}} = \theta^* + \frac{c}{D^*} + o(1) > \theta^*$$

The constant $((1 - \eta)/\eta)^s \sim \eta^{-s}$ diverges, but the η in $C_{\min}/\eta$ compensates, keeping the F1 error bounded.

D.6.2 $\eta \rightarrow 1$ (All Noise)

Corollary D.6. As $\eta \rightarrow 1$: $\theta_{\text{opt}} \rightarrow p_0$ (permissive — flag almost everything). Symmetric to D.5.

D.6.3 $p_0 \rightarrow 0$ (Perfect Experts)

When $p_0 = 0$, $\lambda_0^* \rightarrow \infty$ and $\kappa \rightarrow \infty$. Noise detection becomes trivial — any expert failure definitively indicates noise. The Bahadur-Rao asymptotics degenerate and the bound becomes vacuous, consistent with perfect experts making the problem trivial.

D.7 Constant Optimality Theorem

Theorem D.7 (Exact Constant Optimality of Adaptive SCX). The adaptive threshold SCX detector achieves the minimax lower bound:

$$\lim_{M \rightarrow \infty} e^{M\kappa\sqrt{2\pi M} \cdot (1 - \text{F1}_{\text{SCX}}(\theta_{\text{opt}}))} = \frac{C_{\min}}{\eta}$$

where $C_{\min} = \frac{\eta}{2} \left(\frac{1-\eta}{\eta} \right)^s \cdot \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$ (Lemma E, eq. 45).

Proof. From D.5, the SCX achievable constant is $\left(\frac{1-\eta}{\eta}\right)^s \cdot \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{2\sqrt{\theta^*(1-\theta^*)}}$. Multiply and divide by η :

$$= \frac{1}{\eta} \cdot \frac{\eta}{2} \left(\frac{1-\eta}{\eta}\right)^s \cdot \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}} = \frac{C_{\min}}{\eta}$$

This matches Lemma E’s lower bound exactly. Therefore SCX with adaptive threshold θ_{opt} is **exact constant minimax optimal**. $\square$

Correction (2026-06-27): The original D.4-D.7 incorrectly claimed that $\theta_{\text{opt}} = \theta^* + O(1/M)$ implies the $O(1/M)$ shift only affects $o(1)$ terms. In fact, the $O(1/M)$ shift in θ produces an $O(1)$ **multiplicative factor** $((1-\eta)/\eta)^s$ via the exponential $\exp(-M \cdot \text{KL})$. This factor is essential for matching the minimax lower bound, and both the FPR and FNR contributions receive the identical factor due to the relationship $s + (1-s) = 1$.

FIXED (2026-06-28, DEFECT-08): The self-contradiction in D.4 has been resolved. The corrected derivation now correctly shows: (i) $\theta_{\text{opt}} \neq \theta^*$ but differs by $O(1/M)$; (ii) this $O(1/M)$ shift produces an $O(1)$ multiplicative factor $((1-\eta)/\eta)^s$ in the exponential; (iii) the factor is **identical** in both FPR and FNR contributions — a critical cancellation that makes the adaptive threshold match the minimax lower bound; (iv) Lemma D.4’s “Balance Ratio” and D.5’s exact constant are consistent with Lemma E (canonical $C_{\min}$). **No residual contradiction remains.**

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